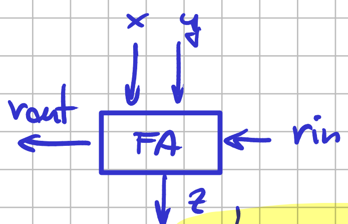
24 ott classroom

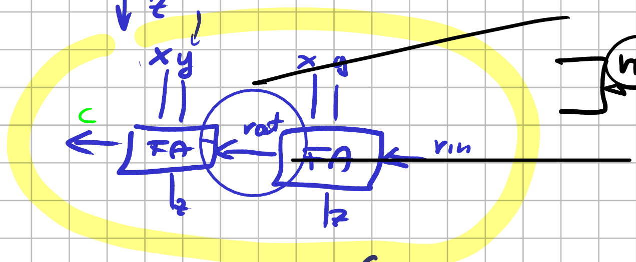
21 nov asm

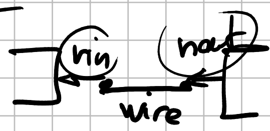
10 dic classroom



fa.v

module fa (output rout,
output z,
input x, input y,
input rin);





module fa2 (output c, output [1:0] z,
input [1:0] x, input [1:0] y, input rin);

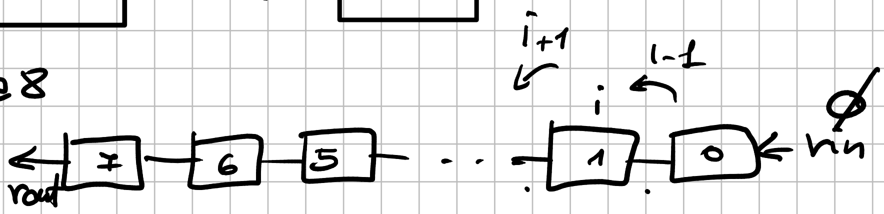
$\boxed{\text{wire}}$ filo;

faN #(16) fa famono (filo, z[0], x[0], y[0], rin);

fa fapiu (c, z[1], x[1], y[1], filo));

$\boxed{\text{generative}}$ → $\boxed{\text{for}}$

fa8

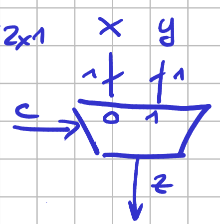


module faN

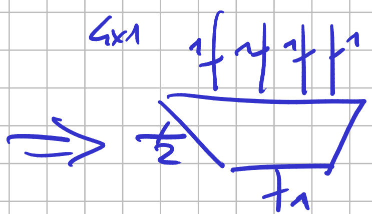
param N = 8 val

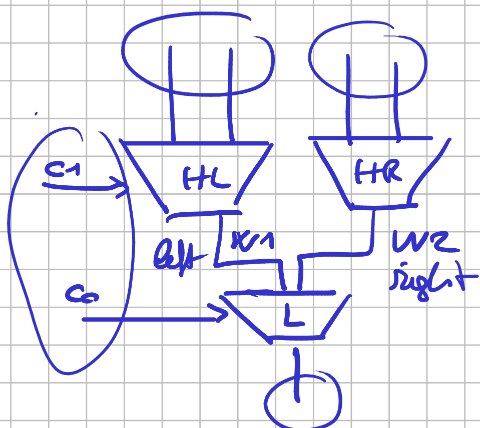
endmodule

MUX2x1



4x1



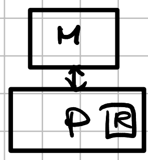


https://www.edaplayground.com/

RETI SEQUENZIALE

Comp$_1$ → Comp$_2$ — . . . → Comp Reg

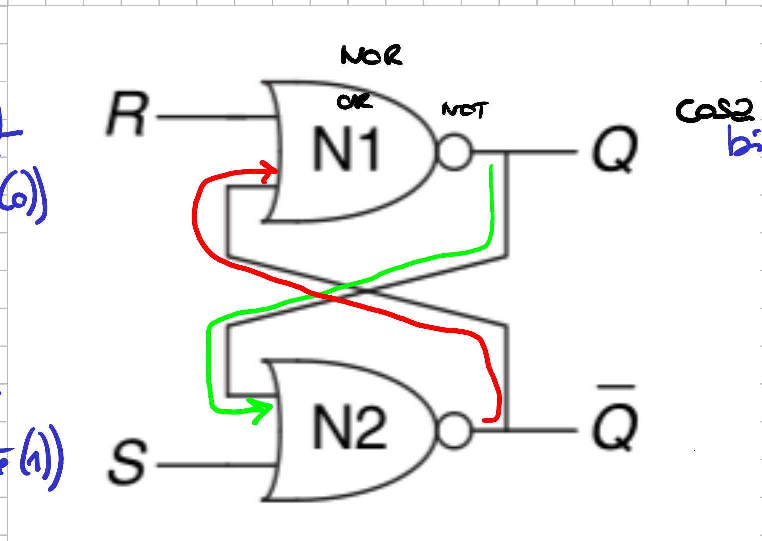
↳ M



SR-LATCH

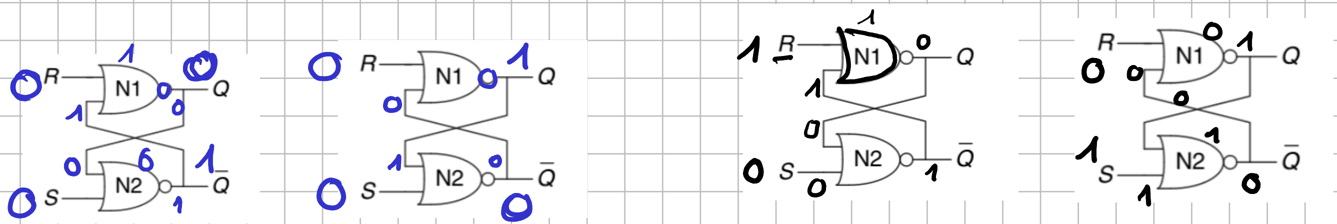
reset (write(0))

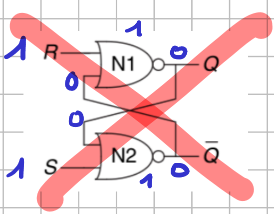
set (write(1))



caso vale il bit che ci vogliamo ricordare

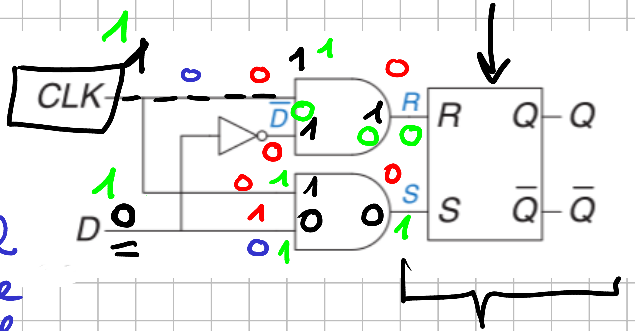
R S	Q $\bar{Q}$
0 0	uscita prec
0 1	1 0
→ 1 0	0 1
1 1	





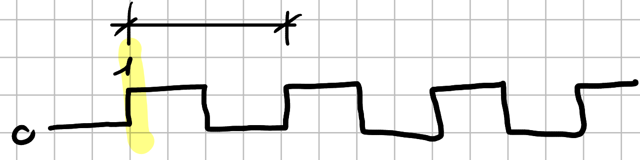
D - latch

SR latch



val che vogliamo ricordare

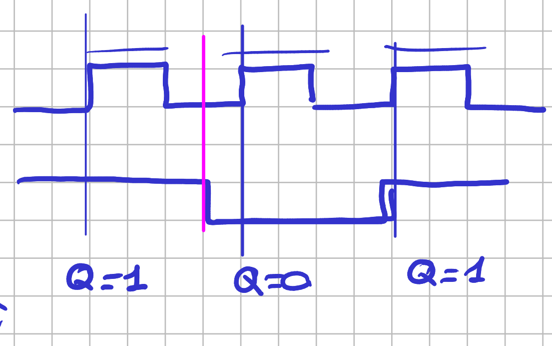
lunghezza del ciclo di clock (τ)



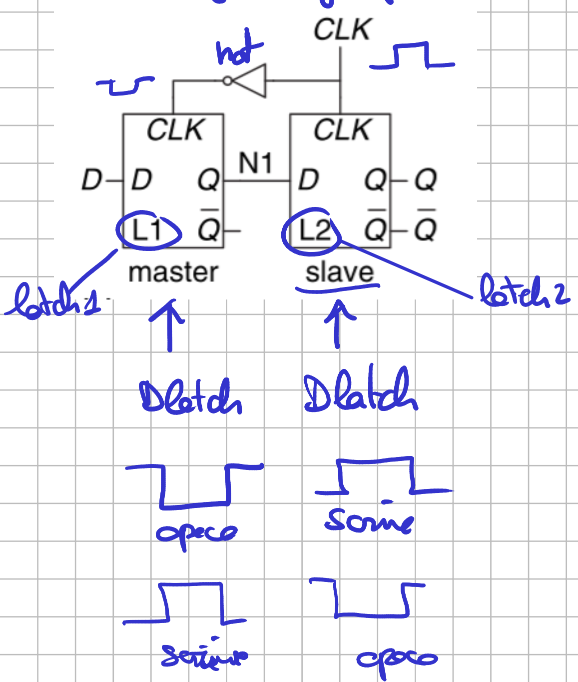
clock

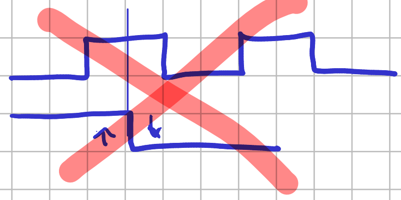
2GHz $\frac{1}{\tau}$

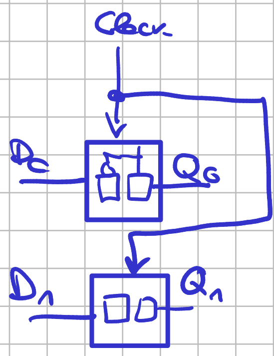
CLK D	R S
0 0	0 0
0 1	0 0
1 0	1 0
1 1	0 1



D flip flop

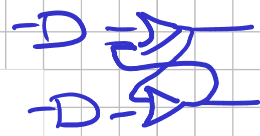






reg 2 bit

2x



10 porte

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